Project ID: 038

Project Title: Semantic Segmentation in Unstructured Environments

Area of Research: Computer Vision, Semantic Segmentation, Unstructured Environment

Problem Statement: Semantic scene understanding is an essential technique in autonomous robot navigation in real-world environments. Existing semantic segmentation methods mainly focus mainly on urban cities. However, they mostly ignore elements that may be present in unstructured environments, e.g., different terrains and vegetations in the mountain places. Fortunately, there is an existing dataset named the Robot Unstructured Ground Driving (RUGD) dataset, which is composed of a set of video sequences collected from a small, unmanned ground robot performing an exploration task in a variety of natural, unstructured environments and semi-urban areas. This project aims to explore an accurate, efficient semantic segmentation model that can be used in unstructured environment using RUGD dataset.

Dataset:

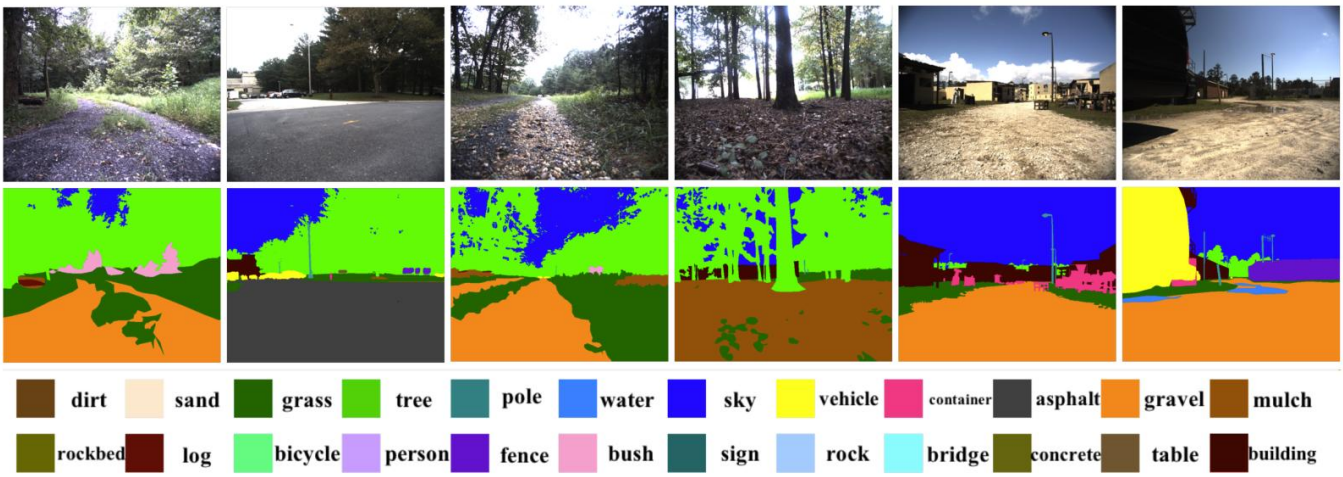

Figure 1 Example ground truth annotations provided in the RUGD dataset. Frames from the video sequences are densely annotated with pixel-wise labels from 24 different visual classes.

Dataset URL: http://rugd.vision/
Dataset paper: https://arxiv.org/html/2404.18750v1

Task: To develop a semantic segmentation model used in unstructured environment using RUGD dataset.

Relevant Papers

[1]. M. Wigness, S. Eum, J. G. Rogers, D. Han and H. Kwon, "A RUGD Dataset for Autonomous Navigation and Visual Perception in Unstructured Outdoor Environments," 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, China, 2019, pp. 5000-5007, doi: 10.1109/IROS40897.2019.8968283.

[2]. Guan, T., Kothandaraman, D., Chandra, R., Sathyamoorthy, A.J., Weerakoon, K.M., & Manocha, D. (2021). GA-Nav: Efficient Terrain Segmentation for Robot Navigation in Unstructured Outdoor Environments. IEEE Robotics and Automation Letters, 7, 8138-8145.

[3]. Wijayathunga, L.; Rassau, A.; Chai, D. Challenges and Solutions for Autonomous Ground Robot Scene Understanding and Navigation in Unstructured Outdoor Environments: A Review. Appl. Sci. 2023, 13, 9877. https://doi.org/10.3390/app13179877